

GRANT COUNTY AP, WV, USA

WMO: 724165

Lat: 38.994N

Lon: 79.144W

Elev: 960

StdP: 14.19

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 03725

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.1	15.7	-5.8	4.2	18.2	0.0	5.7	23.5	26.3	39.3	24.0	37.8	3.4	280	0.416

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	92.9	72.4	90.6	71.9	88.3	71.2	77.6	86.7	75.9	84.9	74.4	82.9	6.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.1	136.5	81.2	73.1	127.6	79.5	72.1	123.2	78.4	41.7	86.0	40.0	85.7	38.5	84.0	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.9	17.8	15.1	DB	2.8	97.0	5.6	3.6	-1.2	99.5	-4.5	101.6	-7.6	103.6	-11.7	106.2	
			WB	1.8	78.6	5.3	2.6	-2.1	80.5	-5.2	82.0	-8.2	83.4	-12.1	85.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.2	34.9	36.8	44.6	55.0	63.6	71.4	74.8	73.6	67.1	56.3	45.5	38.0
	DBStd	16.43	10.89	10.05	9.95	8.75	7.86	5.43	4.94	4.82	6.58	8.34	8.27	8.71
	HDD50	1747	482	384	232	47	3	0	0	0	0	33	182	384
	HDD65	4581	932	791	636	316	125	11	2	3	53	290	586	836
	CDD50	3658	15	14	63	198	426	641	768	731	514	229	46	13
	CDD65	1016	0	0	2	16	83	202	305	269	117	22	0	0
	CDH74	10402	1	5	43	367	915	2021	2928	2518	1269	311	21	3
	CDH80	4014	0	0	4	106	284	751	1250	1050	488	79	2	0
Wind	WSAvg	3.5	4.9	5.2	5.2	4.7	3.0	2.6	2.2	1.8	2.1	3.1	3.5	4.1
Precipitation	PrecAvg	39.5	2.7	2.4	3.6	3.4	4.1	3.6	4.4	3.6	3.5	2.5	2.5	2.8
	PrecMax	60.1	6.2	5.8	6.1	6.7	8.4	6.4	7.7	6.3	9.0	5.6	5.2	4.6
	PrecMin	32.5	1.0	0.7	0.9	1.6	1.7	1.0	1.5	1.9	0.1	0.4	0.7	1.4
	PrecStd	7.0	1.4	1.3	1.4	1.3	1.7	1.5	1.7	1.1	2.5	1.7	1.1	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.7	72.4	79.1	89.5	90.6	93.7	97.4	96.6	93.5	87.2	77.0	70.0
		MCWB	60.1	57.3	58.5	60.9	70.4	72.4	72.4	73.1	69.5	67.9	58.8	59.4
	2%	DB	63.8	64.5	73.1	82.4	87.1	90.8	92.7	91.8	89.9	81.7	71.9	63.3
		MCWB	54.4	52.3	56.3	61.5	69.7	72.7	72.9	73.6	68.6	64.5	57.6	53.5
	5%	DB	57.3	60.6	68.4	78.6	82.6	88.0	90.2	89.8	85.9	77.2	65.8	57.2
		MCWB	49.9	49.8	53.9	60.3	67.4	71.5	72.6	72.5	69.0	63.3	54.6	49.4
	10%	DB	52.0	54.2	63.2	72.9	79.5	84.2	87.6	86.2	81.6	72.2	61.4	52.4
		MCWB	45.4	45.2	52.0	57.9	66.2	70.3	72.3	71.3	67.0	61.3	51.3	46.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.1	60.0	62.7	69.2	76.0	79.5	80.1	79.4	75.9	72.4	63.3	60.5
		MCDB	68.8	72.2	74.4	78.1	84.6	86.1	85.9	89.1	83.2	80.0	71.5	66.8
	2%	WB	56.3	54.8	59.2	64.8	72.8	76.3	77.6	77.2	73.4	68.9	59.5	55.1
		MCDB	61.8	62.6	69.1	75.9	81.7	85.3	86.4	86.9	82.1	76.8	68.4	61.4
	5%	WB	50.4	50.3	56.3	62.4	70.6	74.2	75.8	75.2	71.7	66.0	56.3	51.3
		MCDB	56.8	58.9	66.3	73.5	79.2	82.7	84.9	84.3	80.7	73.6	63.4	56.2
	10%	WB	45.5	45.8	52.5	59.7	68.1	72.3	74.2	73.5	69.7	62.9	53.1	46.4
		MCDB	50.8	52.9	61.5	70.6	76.1	80.7	82.8	81.5	76.8	69.6	59.5	50.6
Mean Daily Temperature Range	5% DB	MDBR	19.0	20.4	22.4	25.7	24.1	23.6	23.3	23.6	25.0	24.4	23.5	19.1
		MCDBR	27.3	30.7	32.2	35.3	30.2	28.0	27.2	28.7	31.6	32.9	33.5	27.8
		MCWBR	18.5	19.9	18.0	17.1	14.8	12.5	10.7	11.7	13.8	17.4	20.5	19.1
	5% WB	MCDBR	25.4	28.3	28.0	28.7	25.3	24.2	23.0	23.4	24.7	26.4	27.0	25.2
		MCWBR	18.8	19.8	17.3	16.0	13.8	12.6	11.1	11.2	13.1	15.3	18.5	19.0
Clear-Sky Solar Irradiance	taub	0.300	0.320	0.349	0.402	0.440	0.493	0.507	0.520	0.411	0.358	0.318	0.299	
	taud	2.474	2.395	2.387	2.267	2.179	2.064	2.046	1.981	2.281	2.432	2.507	2.509	
	Ebn at Noon	279	286	288	278	268	253	248	240	264	269	270	271	
	Edh at Noon	25	31	35	42	47	52	53	55	38	30	24	23	
All-Sky Solar Radiation	RadAvg	599	859	1172	1509	1669	1848	1792	1628	1353	1001	713	526	
	RadStd	58	118	76	126	148	130	107	85	166	117	70	48	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (43 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+205	+110

Nomenclature: See separate page